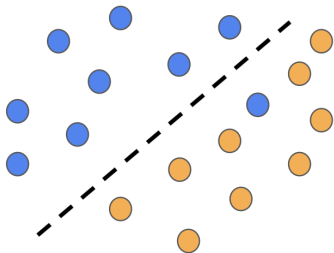


Introduction to Machine Learning

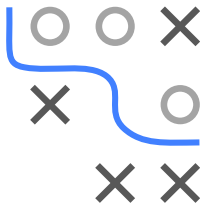
Classification

Linear Classifiers



Learning goals

- Linear classifier
- Linear decision boundaries
- Linear separability



LINEAR CLASSIFIERS I

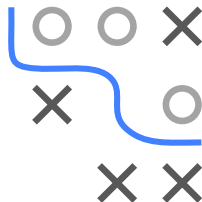
Important subclass of classification models.

Definition: If discriminant(s) $f_k(\mathbf{x})$ can be written as affine linear function(s) (possibly through a rank-preserving, monotone transformation g):

$$g(f_k(\mathbf{x})) = \mathbf{w}_k^\top \mathbf{x} + b_k,$$

we will call the classifier **linear**.

- $\mathbf{w}_k$ and b_k do not necessarily refer to parameters θ_k , although they often coincide; discriminant simply must be writable in an affine-linear way
- reasons for the transformation is that we only care about the position of the decision boundary

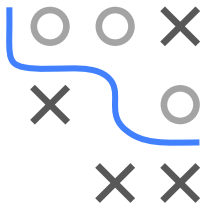
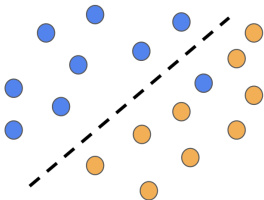


LINEAR DECISION BOUNDARIES I

We can also easily show that the decision boundary between classes i and j is a hyperplane. For every $\mathbf{x}$ where there is a tie in scores:

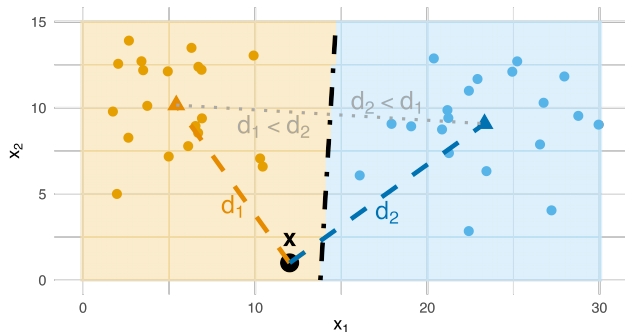
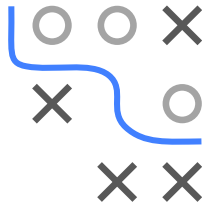
$$\begin{aligned}f_i(\mathbf{x}) &= f_j(\mathbf{x}) \\g(f_i(\mathbf{x})) &= g(f_j(\mathbf{x})) \\ \mathbf{w}_i^\top \mathbf{x} + b_i &= \mathbf{w}_j^\top \mathbf{x} + b_j \\ (\mathbf{w}_i - \mathbf{w}_j)^\top \mathbf{x} + (b_i - b_j) &= 0\end{aligned}$$

This represents a **hyperplane** separating two classes:



EXAMPLE: 2 CLASSES WITH CENTROIDS I

- Model binary problem with centroid μ_k per class as "parameters"
- Don't really care how the centroids are estimated;
could use class means, but the following doesn't depend on it
- Classify point $\mathbf{x}$ by assigning it to class k of nearest centroid



EXAMPLE: 2 CLASSES WITH CENTROIDS I

Let's calculate the decision boundary:

$$d_1 = \|\mathbf{x} - \mu_1\|^2 = \mathbf{x}^\top \mathbf{x} - 2\mathbf{x}^\top \mu_1 + \mu_1^\top \mu_1 = \mathbf{x}^\top \mathbf{x} - 2\mathbf{x}^\top \mu_2 + \mu_2^\top \mu_2 = \|\mathbf{x} - \mu_2\|^2 = d_2$$

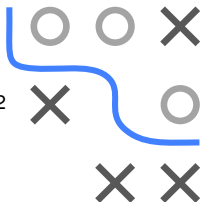
Where d is measured using Euclidean distance. This implies:

$$-2\mathbf{x}^\top \mu_1 + \mu_1^\top \mu_1 = -2\mathbf{x}^\top \mu_2 + \mu_2^\top \mu_2$$

Which simplifies to:

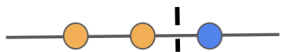
$$2\mathbf{x}^\top (\mu_2 - \mu_1) = \mu_2^\top \mu_2 - \mu_1^\top \mu_1$$

Thus, it's a linear classifier!

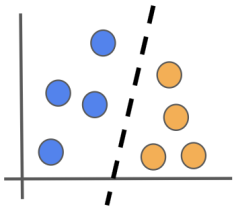


LINEAR SEPARABILITY I

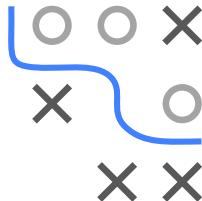
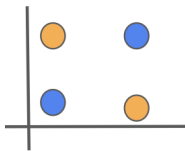
If there exists a linear classifier that perfectly separates the classes of some dataset, the data are called **linearly separable**.



linearly separable

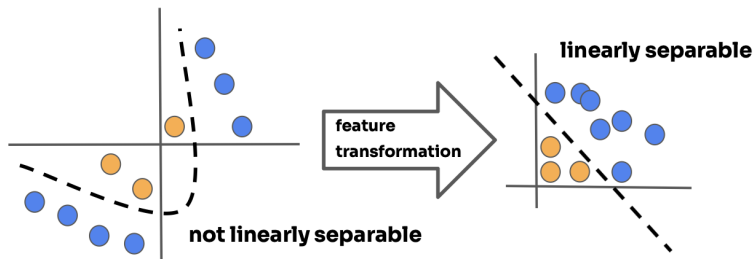
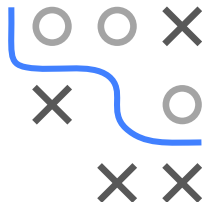


not linearly separable



FEATURE TRANSFORMATIONS I

Note that linear classifiers can represent **non-linear** decision boundaries in the original input space if we use derived features like higher order interactions, polynomial features, etc.



Here we used absolute values to find suitable derived features.